

Programming II

Recursion

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Recursion

- Recursion in java is a process in which a method calls itself continuously.
- A method in java that calls itself is called recursive method.
- It makes the code compact but complex to understand.

Recursive Method:

- Divide the problem into smaller problems until we reach a simple problem that we can solve.
- A recursive method is one that invokes itself.
- With each call of the method the problem becomes easier.

Recursive Method:

- Any problem with loop can be solved in both ways:
 - Recursively
 - Non-recursively
- The problem without loop can't be solved using recursion.

Example:

- Many mathematical functions are defined using recursion.

Example: Factorial problem

- $1! = 1$
- $n! = n \times (n-1)! , n > 0$
 $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$

Main Components Of Recursive Method:

A recursive method consist of the following:

- **Base case (Stopping condition):** The simple problem which is the if condition which means no need to call the function again.
- **The process and the recursive part:** calling the same function again

The factorial Example:

- Write a recursive method to find the factorial of n:

```
if(n==1) n!=1 // Base case  
else  
n!=n*(n-1)! // Recursive call
```

Recursive Steps For Executing Recursive Method:

```
public static int factorial(int n) {  
    if (n == 1)  
        return 1;  
    else  
        return n * factorial(n - 1);  
}
```

At n=5
1-5*?
2-4*?
3-3*?
4-2*?
5- return 1

Program:

```
public class Main
{
    public static void main(String[] args) {
        System.out.println(fact(4));
    }
    public static int fact (int n)
    { if ( n == 1)
        return 1;
      else
        return n*fact(n-1);
    }
}
```

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Example:

What is the output of the following program?

```
public class Test {  
    public static void main(String[] args) {  
        System.out.println(  
            "Sum is " + summation (5));  
    }  
  
    public static int summation (int n) {  
        if (n == 1)  
            return 1;  
        else  
            return n + summation (n - 1);  
    }  
}
```

At n=5

1-5+?(10) =15

final return

2-4+? (6)=10

3-3+? (3)=6

4-2+ (1)=3

return 1

Example

- **Write a recursive method to calculate 2^n for a positive integer n.**

$$2^5 = 2 \times 2^4$$

Stopping Case at $n=0$ because $2^0=1$

```
public static int pow2(int x){  
    if (x==0)  
        return 1;  
    else  
        return 2*pow2(x-1);  
}
```

Program:

```
import java.util.Scanner;
public class Main
{
    public static void main(String[] args) {
        Scanner input = new Scanner (System.in);
        System.out.println ("Enter a number");
        int y = input.nextInt();
        System.out.println("2^"+y+" = "+pow2(y));
    }
    public static int pow2(int x){
        if (x==0)
            return 1;
        else
            return 2*pow2(x-1);
    }
}
```

```
Enter a number
5
2^5 = 32
```